

CPSC 304 Project Cover Page

Milestone #: 2

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Group Number: 38

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above. (In the case of Project Milestone 0, the main purpose of this page is for you to let us know your e-mail address, and then let us assign you to a TA for your project supervisor.)

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

Project Summary

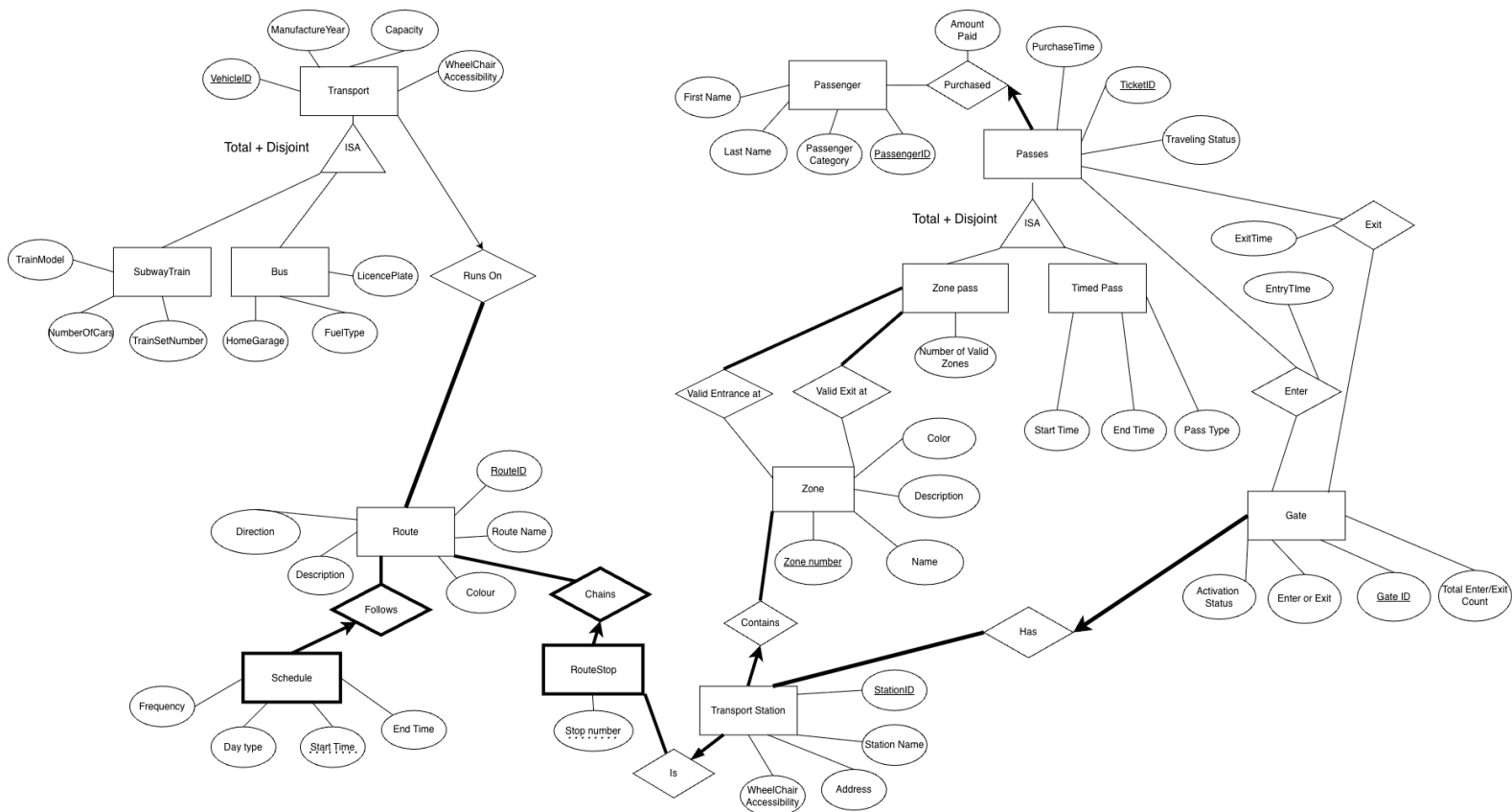
Our project builds a database for managing public transit. It covers buses, subway trains, passengers, transit passes, routes, stations, gates, and fare zones. The system tracks vehicles, stores route and schedule details, checks whether a passenger's transit pass is valid, and logs every time someone enters or exits a station. We built the database using normalized relational design, so the data stays accurate and the system runs efficiently even as the number of records grows.

Database and Programming Language

Our frontend runs on standard HTML and JavaScript. The backend runs on Node.js. To connect to the department's Oracle database, we'll use the `oracledb` driver and write our own SQL queries directly, without an ORM.

ER Diagram

We're using the same ER diagram from Milestone 1 for this milestone. Our project mentor didn't request any changes or give feedback on the design, so we kept the original diagram as is. It's the basis for the relational schema and SQL implementation we built this time around.



Transport Database Relational Schema

Relational Schema

1. Transport

Transport(VehicleID: char(10), ManufactureYear: number(4), Capacity: number(4), WheelchairAccessibility: char(1))

PK: VehicleID

CK: None other than the primary key

FK: None

Constraints:

- VehicleID cannot be null.
- ManufactureYear cannot be null and must be greater than or equal to 1900.
- Capacity cannot be null and must be greater than 0.
- WheelchairAccessibility cannot be null and must be either “Y” or “N”.

2. Bus

Bus(VehicleID: char(10), LicencePlate: varchar2(10), HomeGarage: varchar2(60), FuelType: varchar2(25))

PK: VehicleID

CK: LicencePlate

FK: VehicleID references Transport(VehicleID)

Constraints:

- VehicleID cannot be null.
- LicencePlate cannot be null and must be unique.
- LicencePlate must contain between 2 and 10 uppercase letters and numbers.

3. TrainModelInfo

TrainModelInfo(TrainModel: varchar2(40), NumberOfCars: number(2))

PK: TrainModel

CK: None other than the primary key

FK: None

Constraints:

- TrainModel cannot be null.
- NumberOfCars cannot be null and must be greater than 0.

4. SubwayTrain

SubwayTrain(VehicleID: char(10), TrainSetNumber: varchar2(20), TrainModel: varchar2(40))

PK: VehicleID

CK: TrainSetNumber

FK:

- VehicleID references Transport(VehicleID).
- TrainModel references TrainModelInfo(TrainModel).

Constraints:

- VehicleID cannot be null.
- TrainSetNumber cannot be null and must be unique.
- TrainSetNumber may contain only uppercase letters, digits, and hyphens.
- TrainModel cannot be null.

5. Passenger

Passenger(PassengerID: char(10), FirstName: varchar2(20), LastName: varchar2(20), PassengerCategory: varchar2(20))

PK: PassengerID

CK: None other than the primary key

FK: None

Constraints:

- PassengerID cannot be null.
- FirstName cannot be null.

- LastName cannot be null.
- PassengerCategory cannot be null.
- PassengerCategory must be one of: Adult, Child, Senior, or Student.

6. Passes

Passes(TicketID: char(10), PassengerID: char(10), TravellingStatus: varchar2(20), AmountPaid: number(8,2), PurchaseTime: timestamp)

PK: TicketID

CK: None other than the primary key

FK: PassengerID references Passenger(PassengerID)

Constraints:

- TicketID cannot be null.
- PassengerID cannot be null.
- TravellingStatus cannot be null.
- TravellingStatus must be one of: Active, Inactive, Expired, or Suspended.
- AmountPaid cannot be null and must be greater than or equal to 0.
- PurchaseTime cannot be null and defaults to the current timestamp.

7. ZonePass

ZonePass(TicketID: char(10), NumberOfValidZones: number(2))

PK: TicketID

CK: None other than the primary key

FK: TicketID references Passes(TicketID)

Constraints:

- TicketID cannot be null.
- NumberOfValidZones cannot be null and must be greater than 0.

8. TimedPass

TimedPass(TicketID: char(10), StartTime: timestamp, EndTime: timestamp, PassType: varchar2(20))

PK: TicketID

CK: None other than the primary key

FK: TicketID references Passes(TicketID)

Constraints:

- TicketID cannot be null.
- StartTime cannot be null.
- EndTime cannot be null.
- EndTime must be later than StartTime.
- PassType cannot be null.
- PassType must be one of: Daily, Weekly, or Monthly.

9. Route

Route(RouteID: char(10), RouteName: varchar2(60), Direction: varchar2(20), RouteDescription: varchar2(200), Colour: varchar2(20))

PK: RouteID

CK: None other than the primary key

FK: None

Constraints:

- RouteID cannot be null.
- RouteName cannot be null.
- Direction cannot be null.

10. Schedule

Schedule(RouteID: char(10), StartTime: timestamp, EndTime: timestamp, Frequency: number(4), DayType: varchar2(20))

PK: (RouteID, StartTime, DayType)

CK: None other than the primary key

FK: RouteID references Route(RouteID)

Constraints:

- RouteID cannot be null.
- StartTime cannot be null.
- EndTime cannot be null.
- EndTime must be later than StartTime.
- Frequency cannot be null and must be greater than 0.

- DayType cannot be null.
- DayType must be one of: Weekday, Weekend, or Holiday.

11. Zone

Zone(ZoneNumber: number(3), ZoneName: varchar2(50), Colour: varchar2(20), ZoneDescription: varchar2(200))

PK: ZoneNumber

CK: None other than the primary key

FK: None

Constraints:

- ZoneNumber cannot be null and must be greater than 0.
- ZoneName cannot be null.

12. TransportStation

TransportStation(StationID: char(10), ZoneNumber: number(3), StationName: varchar2(80), Address: varchar2(150), WheelchairAccessibility: char(1))

PK: StationID

CK: None other than the primary key

FK: ZoneNumber references Zone(ZoneNumber)

Constraints:

- StationID cannot be null.
- ZoneNumber cannot be null.
- StationName cannot be null.
- Address cannot be null.
- WheelchairAccessibility cannot be null and must be either “Y” or “N”.

13. RouteStop

RouteStop(RouteID: char(10), StopNumber: number(4), StationID: char(10))

PK: (RouteID, StopNumber)

CK: None other than the primary key

FK:

- RouteID references Route(RouteID).

- StationID references TransportStation(StationID).

Constraints:

- RouteID cannot be null.
- StationID cannot be null.
- StopNumber cannot be null and must be greater than 0.

14. Gate

Gate(GateID: char(10), StationID: char(10), ActivationStatus: varchar2(20), EnterOrExit: varchar2(10), TotalEnterExitCount: number(10))

PK: GateID

CK: None other than the primary key

FK: StationID references TransportStation(StationID)

Constraints:

- GateID cannot be null.
- StationID cannot be null.
- ActivationStatus cannot be null.
- ActivationStatus must be one of: Active or Inactive.
- EnterOrExit cannot be null.
- EnterOrExit must be one of: Enter or Exit.
- TotalEnterExitCount cannot be null, defaults to 0, and cannot be negative.

Relations Derived from Many-to-Many Relationships

15. RunsOn

RunsOn(VehicleID: char(10), RouteID: char(10))

PK: (VehicleID, RouteID)

CK: None other than the primary key

FK:

- VehicleID references Transport(VehicleID).
- RouteID references Route(RouteID).

Constraints:

- VehicleID cannot be null.
- RouteID cannot be null.

16. ValidEntranceAt

ValidEntranceAt(TicketID: char(10), ZoneNumber: number(3))

PK: (TicketID, ZoneNumber)

CK: None other than the primary key

FK:

- TicketID references ZonePass(TicketID).
- ZoneNumber references Zone(ZoneNumber).

Constraints:

- TicketID cannot be null.
- ZoneNumber cannot be null.

17. ValidExitAt

ValidExitAt(TicketID: char(10), ZoneNumber: number(3))

PK: (TicketID, ZoneNumber)

CK: None other than the primary key

FK:

- TicketID references ZonePass(TicketID).
- ZoneNumber references Zone(ZoneNumber).

Constraints:

- TicketID cannot be null.
- ZoneNumber cannot be null.

18. PassEnter

PassEnter(TicketID: char(10), GateID: char(10), EntryTime: timestamp)

PK: (TicketID, GateID, EntryTime)

CK: None other than the primary key

FK:

- TicketID references Passes(TicketID).

- GateID references Gate(GateID).

Constraints:

- TicketID cannot be null.
- GateID cannot be null.
- EntryTime cannot be null and defaults to the current timestamp.

19. PassExit

PassExit(TicketID: char(10), GateID: char(10), ExitTime: timestamp)

PK: (TicketID, GateID, ExitTime)

CK: None other than the primary key

FK:

- TicketID references Passes(TicketID).
- GateID references Gate(GateID).

Constraints:

- TicketID cannot be null.
- GateID cannot be null.
- ExitTime cannot be null and defaults to the current timestamp.

Functional Dependencies

1. Transport

$VehicleID \rightarrow ManufactureYear, Capacity, WheelchairAccessibility$

2. Bus

$VehicleID \rightarrow LicencePlate, HomeGarage, FuelType$

$LicencePlate \rightarrow VehicleID, HomeGarage, FuelType$

3. TrainModelInfo

$TrainModel \rightarrow NumberOfCars$

4. SubwayTrain

$VehicleID \rightarrow TrainSetNumber, TrainModel$

$TrainSetNumber \rightarrow VehicleID, TrainModel$

5. Passenger

$PassengerID \rightarrow FirstName, LastName, PassengerCategory$

6. Passes

$TicketID \rightarrow PassengerID, TravellingStatus, AmountPaid, PurchaseTime$

7. ZonePass

$TicketID \rightarrow NumberOfValidZones$

8. TimedPass

$TicketID \rightarrow StartTime, EndTime, PassType$

9. Route

$RouteID \rightarrow RouteName, Direction, RouteDescription, Colour$

10. Schedule

$RouteID, StartTime, DayType \rightarrow EndTime, Frequency$

11. Zone

$ZoneNumber \rightarrow ZoneName, Colour, ZoneDescription$

12. TransportStation

$StationID \rightarrow ZoneNumber, StationName, Address, WheelchairAccessibility$

13. **RouteStop**

$RouteID, StopNumber \rightarrow StationID$

14. **Gate**

$GateID \rightarrow StationID, ActivationStatus, EnterOrExit, TotalEnterExitCount$

The relations RunsOn, ValidEntranceAt, ValidExitAt, PassEnter, and PassExit contain only key attributes.

Normalization of SubwayTrain

Original relation:

SubwayTrain(VehicleID, TrainModel, NumberOfCars, TrainSetNumber)

Candidate keys:

- VehicleID
- TrainSetNumber

Functional dependencies:

$VehicleID \rightarrow TrainModel, NumberOfCars, TrainSetNumber$

$TrainSetNumber \rightarrow VehicleID, TrainModel, NumberOfCars$

$TrainModel \rightarrow NumberOfCars$

Decomposition:

TrainModelInfo(TrainModel, NumberOfCars)

SubwayTrain(VehicleID, TrainSetNumber, TrainModel)

The decomposition is lossless because the common attribute, TrainModel, is the primary key of TrainModelInfo. The dependency

$TrainModel \rightarrow NumberOfCars$

is preserved in TrainModelInfo. The resulting relations are in BCNF.

Implementation Note

Subtype disjointness, subtype completeness, and minimum participation constraints that cannot be expressed using the current SQL DDL will be enforced later using application logic, triggers, or assertions.

SQL DDL Statements

DDL.sql - Table Creation Statements

```
1  -- 1. Transport
2  CREATE TABLE Transport (
3      VehicleID CHAR(10) PRIMARY KEY,
4      ManufactureYear NUMBER(4) NOT NULL CHECK (ManufactureYear >= 1900),
5      Capacity NUMBER(4) NOT NULL CHECK (Capacity > 0),
6      WheelchairAccessibility CHAR(1) NOT NULL CHECK (WheelchairAccessibility IN ('Y', 'N'))
7  );
8
9  -- 2. Bus
10 CREATE TABLE Bus (
11     VehicleID CHAR(10) PRIMARY KEY,
12     LicencePlate VARCHAR2(10) UNIQUE NOT NULL,
13     HomeGarage VARCHAR2(60),
14     FuelType VARCHAR2(25),
15     FOREIGN KEY (VehicleID) REFERENCES Transport(VehicleID)
16 );
17
18 -- 3. TrainModelInfo
19 CREATE TABLE TrainModelInfo (
20     TrainModel VARCHAR2(40) PRIMARY KEY,
21     NumberOfCars NUMBER(2) NOT NULL CHECK (NumberOfCars > 0)
22 );
23
24 -- 4. SubwayTrain
25 CREATE TABLE SubwayTrain (
26     VehicleID CHAR(10) PRIMARY KEY,
27     TrainSetNumber VARCHAR2(20) UNIQUE NOT NULL,
28     TrainModel VARCHAR2(40) NOT NULL,
29     FOREIGN KEY (VehicleID) REFERENCES Transport(VehicleID),
30     FOREIGN KEY (TrainModel) REFERENCES TrainModelInfo(TrainModel)
31 );
32
33 -- 5. Passenger
34 CREATE TABLE Passenger (
35     PassengerID CHAR(10) PRIMARY KEY,
36     FirstName VARCHAR2(20) NOT NULL,
37     LastName VARCHAR2(20) NOT NULL,
38     PassengerCategory VARCHAR2(20) NOT NULL CHECK (PassengerCategory IN ('Adult', 'Child', 'Senior', 'Student'))
39 );
40
41 -- 6. Passes
42 CREATE TABLE Passes (
43     TicketID CHAR(10) PRIMARY KEY,
44     PassengerID CHAR(10) NOT NULL,
45     TravellingStatus VARCHAR2(20) NOT NULL CHECK (TravellingStatus IN ('Active', 'Inactive', 'Expired', 'Suspended'
46 )),
47     AmountPaid NUMBER(8,2) NOT NULL CHECK (AmountPaid >= 0),
48     PurchaseTime TIMESTAMP DEFAULT CURRENT_TIMESTAMP NOT NULL,
49     FOREIGN KEY (PassengerID) REFERENCES Passenger(PassengerID)
50 );
51
52 -- 7. ZonePass
53 CREATE TABLE ZonePass (
54     TicketID CHAR(10) PRIMARY KEY,
55     NumberOfValidZones NUMBER(2) NOT NULL CHECK (NumberOfValidZones > 0),
56     FOREIGN KEY (TicketID) REFERENCES Passes(TicketID)
57 );
58
59 -- 8. TimedPass
60 CREATE TABLE TimedPass (
61     TicketID CHAR(10) PRIMARY KEY,
62     StartTime TIMESTAMP NOT NULL,
63     EndTime TIMESTAMP NOT NULL,
64     PassType VARCHAR2(20) NOT NULL CHECK (PassType IN ('Daily', 'Weekly', 'Monthly')),
65     FOREIGN KEY (TicketID) REFERENCES Passes(TicketID),
66     CONSTRAINT chk_timedpass_time CHECK (EndTime > StartTime)
67 );
68
69 -- 9. Route
70 CREATE TABLE Route (
71     RouteID CHAR(10) PRIMARY KEY,
```

DDL.sql - Table Creation Statements (continued)

```
71     RouteName VARCHAR2(60) NOT NULL,
72     Direction VARCHAR2(20) NOT NULL,
73     RouteDescription VARCHAR2(200),
74     Colour VARCHAR2(20)
75 );
76
77 -- 10. Schedule
78 CREATE TABLE Schedule (
79     RouteID CHAR(10),
80     StartTime TIMESTAMP,
81     DayType VARCHAR2(20) CHECK (DayType IN ('Weekday', 'Weekend', 'Holiday')),
82     EndTime TIMESTAMP NOT NULL,
83     Frequency NUMBER(4) NOT NULL CHECK (Frequency > 0),
84     PRIMARY KEY (RouteID, StartTime, DayType),
85     FOREIGN KEY (RouteID) REFERENCES Route(RouteID),
86     CONSTRAINT chk_schedule_time CHECK (EndTime > StartTime)
87 );
88
89 -- 11. Zone
90 CREATE TABLE Zone (
91     ZoneNumber NUMBER(3) PRIMARY KEY CHECK (ZoneNumber > 0),
92     ZoneName VARCHAR2(50) NOT NULL,
93     Colour VARCHAR2(20),
94     ZoneDescription VARCHAR2(200)
95 );
96
97 -- 12. TransportStation
98 CREATE TABLE TransportStation (
99     StationID CHAR(10) PRIMARY KEY,
100    ZoneNumber NUMBER(3) NOT NULL,
101    StationName VARCHAR2(80) NOT NULL,
102    Address VARCHAR2(150) NOT NULL,
103    WheelchairAccessibility CHAR(1) NOT NULL CHECK (WheelchairAccessibility IN ('Y', 'N')),
104    FOREIGN KEY (ZoneNumber) REFERENCES Zone(ZoneNumber)
105 );
106
107 -- 13. RouteStop
108 CREATE TABLE RouteStop (
109     RouteID CHAR(10),
110     StopNumber NUMBER(4) CHECK (StopNumber > 0),
111     StationID CHAR(10) NOT NULL,
112     PRIMARY KEY (RouteID, StopNumber),
113     FOREIGN KEY (RouteID) REFERENCES Route(RouteID),
114     FOREIGN KEY (StationID) REFERENCES TransportStation(StationID)
115 );
116
117 -- 14. Gate
118 CREATE TABLE Gate (
119     GateID CHAR(10) PRIMARY KEY,
120     StationID CHAR(10) NOT NULL,
121     ActivationStatus VARCHAR2(20) NOT NULL CHECK (ActivationStatus IN ('Active', 'Inactive')),
122     EnterOrExit VARCHAR2(10) NOT NULL CHECK (EnterOrExit IN ('Enter', 'Exit')),
123     TotalEnterExitCount NUMBER(10) DEFAULT 0 NOT NULL CHECK (TotalEnterExitCount >= 0),
124     FOREIGN KEY (StationID) REFERENCES TransportStation(StationID)
125 );
126
127 -- 15. RunsOn
128 CREATE TABLE RunsOn (
129     VehicleID CHAR(10),
130     RouteID CHAR(10),
131     PRIMARY KEY (VehicleID, RouteID),
132     FOREIGN KEY (VehicleID) REFERENCES Transport(VehicleID),
133     FOREIGN KEY (RouteID) REFERENCES Route(RouteID)
134 );
135
136 -- 16. ValidEntranceAt
137 CREATE TABLE ValidEntranceAt (
138     TicketID CHAR(10),
139     ZoneNumber NUMBER(3),
140     PRIMARY KEY (TicketID, ZoneNumber),
141     FOREIGN KEY (TicketID) REFERENCES ZonePass(TicketID),
```

DDL.sql - Table Creation Statements (continued)

```
142     FOREIGN KEY (ZoneNumber) REFERENCES Zone(ZoneNumber)
143 );
144
145 -- 17. ValidExitAt
146 CREATE TABLE ValidExitAt (
147     TicketID CHAR(10),
148     ZoneNumber NUMBER(3),
149     PRIMARY KEY (TicketID, ZoneNumber),
150     FOREIGN KEY (TicketID) REFERENCES ZonePass(TicketID),
151     FOREIGN KEY (ZoneNumber) REFERENCES Zone(ZoneNumber)
152 );
153
154 -- 18. PassEnter
155 CREATE TABLE PassEnter (
156     TicketID CHAR(10),
157     GateID CHAR(10),
158     EntryTime TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
159     PRIMARY KEY (TicketID, GateID, EntryTime),
160     FOREIGN KEY (TicketID) REFERENCES Passes(TicketID),
161     FOREIGN KEY (GateID) REFERENCES Gate(GateID)
162 );
163
164 -- 19. PassExit
165 CREATE TABLE PassExit (
166     TicketID CHAR(10),
167     GateID CHAR(10),
168     ExitTime TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
169     PRIMARY KEY (TicketID, GateID, ExitTime),
170     FOREIGN KEY (TicketID) REFERENCES Passes(TicketID),
171     FOREIGN KEY (GateID) REFERENCES Gate(GateID)
172 );
```


ddl2.sql - Data Insertion Statements

```
1  -- 1. Transport
2  INSERT INTO Transport VALUES ('V001', 2015, 60, 'Y');
3  INSERT INTO Transport VALUES ('V002', 2018, 45, 'Y');
4  INSERT INTO Transport VALUES ('V003', 2010, 80, 'N');
5  INSERT INTO Transport VALUES ('V004', 2021, 120, 'Y');
6  INSERT INTO Transport VALUES ('V005', 2020, 100, 'Y');
7
8  -- 2. Bus
9  INSERT INTO Bus VALUES ('V001', 'AB123C', 'North Garage', 'Diesel');
10 INSERT INTO Bus VALUES ('V002', 'XY987Z', 'South Garage', 'Electric');
11 INSERT INTO Bus VALUES ('V003', 'BC456D', 'East Garage', 'Hybrid');
12 INSERT INTO Bus VALUES ('V004', 'MN345P', 'West Garage', 'Electric');
13 INSERT INTO Bus VALUES ('V005', 'PQ765R', 'North Garage', 'Diesel');
14
15 -- 3. TrainModelInfo
16 INSERT INTO TrainModelInfo VALUES ('Mark I', 4);
17 INSERT INTO TrainModelInfo VALUES ('Mark II', 2);
18 INSERT INTO TrainModelInfo VALUES ('Mark III', 4);
19 INSERT INTO TrainModelInfo VALUES ('Mark V', 6);
20 INSERT INTO TrainModelInfo VALUES ('Alstom', 5);
21
22 -- 4. SubwayTrain
23 INSERT INTO SubwayTrain VALUES ('V001', 'TS-100', 'Mark I');
24 INSERT INTO SubwayTrain VALUES ('V002', 'TS-101', 'Mark II');
25 INSERT INTO SubwayTrain VALUES ('V003', 'TS-102', 'Mark III');
26 INSERT INTO SubwayTrain VALUES ('V004', 'TS-103', 'Mark V');
27 INSERT INTO SubwayTrain VALUES ('V005', 'TS-104', 'Alstom');
28
29 -- 5. Passenger
30 INSERT INTO Passenger VALUES ('P001', 'John', 'Doe', 'Adult');
31 INSERT INTO Passenger VALUES ('P002', 'Jane', 'Smith', 'Student');
32 INSERT INTO Passenger VALUES ('P003', 'Alice', 'Johnson', 'Senior');
33 INSERT INTO Passenger VALUES ('P004', 'Bob', 'Brown', 'Child');
34 INSERT INTO Passenger VALUES ('P005', 'Charlie', 'Davis', 'Adult');
35
36 -- 6. Passes
37 INSERT INTO Passes VALUES ('T001', 'P001', 'Active', 105.00, CURRENT_TIMESTAMP);
38 INSERT INTO Passes VALUES ('T002', 'P002', 'Active', 55.00, CURRENT_TIMESTAMP);
39 INSERT INTO Passes VALUES ('T003', 'P003', 'Expired', 45.00, CURRENT_TIMESTAMP);
40 INSERT INTO Passes VALUES ('T004', 'P004', 'Active', 30.00, CURRENT_TIMESTAMP);
41 INSERT INTO Passes VALUES ('T005', 'P005', 'Inactive', 0.00, CURRENT_TIMESTAMP);
42
43 -- 7. ZonePass
44 INSERT INTO ZonePass VALUES ('T001', 3);
45 INSERT INTO ZonePass VALUES ('T002', 1);
46 INSERT INTO ZonePass VALUES ('T003', 2);
47 INSERT INTO ZonePass VALUES ('T004', 1);
48 INSERT INTO ZonePass VALUES ('T005', 2);
49
50 -- 8. TimedPass
51 INSERT INTO TimedPass VALUES ('T001', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP + INTERVAL '1' MONTH, 'Monthly');
52 INSERT INTO TimedPass VALUES ('T002', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP + INTERVAL '7' DAY, 'Weekly');
53 INSERT INTO TimedPass VALUES ('T003', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP + INTERVAL '1' DAY, 'Daily');
54 INSERT INTO TimedPass VALUES ('T004', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP + INTERVAL '1' MONTH, 'Monthly');
55 INSERT INTO TimedPass VALUES ('T005', CURRENT_TIMESTAMP, CURRENT_TIMESTAMP + INTERVAL '7' DAY, 'Weekly');
56
57 -- 9. Route
58 INSERT INTO Route VALUES ('R001', '99 B-Line', 'Eastbound', 'Express to UBC', 'Orange');
59 INSERT INTO Route VALUES ('R002', 'Expo Line', 'Westbound', 'Downtown to Surrey', 'Blue');
60 INSERT INTO Route VALUES ('R003', 'Canada Line', 'Northbound', 'Richmond to Downtown', 'Light Blue');
61 INSERT INTO Route VALUES ('R004', 'Millennium', 'Eastbound', 'VCC to Lafarge', 'Yellow');
62 INSERT INTO Route VALUES ('R005', '49', 'Westbound', 'Metrotown to UBC', 'Red');
63
64 -- 10. Schedule
65 INSERT INTO Schedule VALUES ('R001', TO_TIMESTAMP('08:00:00', 'HH24:MI:SS'), 'Weekday', TO_TIMESTAMP('22:00:00', 'HH24:MI:SS'), 5);
66 INSERT INTO Schedule VALUES ('R002', TO_TIMESTAMP('06:00:00', 'HH24:MI:SS'), 'Weekday', TO_TIMESTAMP('01:00:00', 'HH24:MI:SS'), 3);
67 INSERT INTO Schedule VALUES ('R003', TO_TIMESTAMP('07:00:00', 'HH24:MI:SS'), 'Weekend', TO_TIMESTAMP('23:00:00', 'HH24:MI:SS'), 10);
```

ddl2.sql - Data Insertion Statements (continued)

```
68 INSERT INTO Schedule VALUES ('R004', TO_TIMESTAMP('09:00:00', 'HH24:MI:SS'), 'Holiday', TO_TIMESTAMP('20:00:00',
   'HH24:MI:SS'), 15);
69 INSERT INTO Schedule VALUES ('R005', TO_TIMESTAMP('05:30:00', 'HH24:MI:SS'), 'Weekday', TO_TIMESTAMP('23:30:00',
   'HH24:MI:SS'), 10);
70
71 -- 11. Zone
72 INSERT INTO Zone VALUES (1, 'Vancouver', 'Green', 'Central Hub');
73 INSERT INTO Zone VALUES (2, 'Burnaby/Richmond', 'Yellow', 'Inner Suburbs');
74 INSERT INTO Zone VALUES (3, 'Surrey/Langley', 'Red', 'Outer Suburbs');
75 INSERT INTO Zone VALUES (4, 'North Shore', 'Blue', 'Mountains');
76 INSERT INTO Zone VALUES (5, 'Delta', 'Orange', 'South of Fraser');
77
78 -- 12. TransportStation
79 INSERT INTO TransportStation VALUES ('S001', 1, 'Waterfront', '601 W Cordova St', 'Y');
80 INSERT INTO TransportStation VALUES ('S002', 2, 'Metrotown', '6300 Central Blvd', 'Y');
81 INSERT INTO TransportStation VALUES ('S003', 3, 'Surrey Central', '10275 City Pkwy', 'Y');
82 INSERT INTO TransportStation VALUES ('S004', 1, 'UBC Exchange', '6131 Student Union Blvd', 'Y');
83 INSERT INTO TransportStation VALUES ('S005', 2, 'Richmond-Brighouse', '6971 No 3 Rd', 'Y');
84
85 -- 13. RouteStop
86 INSERT INTO RouteStop VALUES ('R001', 1, 'S004');
87 INSERT INTO RouteStop VALUES ('R002', 1, 'S001');
88 INSERT INTO RouteStop VALUES ('R002', 15, 'S003');
89 INSERT INTO RouteStop VALUES ('R003', 1, 'S005');
90 INSERT INTO RouteStop VALUES ('R005', 1, 'S002');
91
92 -- 14. Gate
93 INSERT INTO Gate VALUES ('G001', 'S001', 'Active', 'Enter', 15000);
94 INSERT INTO Gate VALUES ('G002', 'S001', 'Active', 'Exit', 14500);
95 INSERT INTO Gate VALUES ('G003', 'S002', 'Active', 'Enter', 8000);
96 INSERT INTO Gate VALUES ('G004', 'S003', 'Inactive', 'Exit', 200);
97 INSERT INTO Gate VALUES ('G005', 'S004', 'Active', 'Enter', 5000);
98
99 -- 15. RunsOn
100 INSERT INTO RunsOn VALUES ('V001', 'R001');
101 INSERT INTO RunsOn VALUES ('V002', 'R002');
102 INSERT INTO RunsOn VALUES ('V003', 'R003');
103 INSERT INTO RunsOn VALUES ('V004', 'R004');
104 INSERT INTO RunsOn VALUES ('V005', 'R005');
105
106 -- 16. ValidEntranceAt
107 INSERT INTO ValidEntranceAt VALUES ('T001', 1);
108 INSERT INTO ValidEntranceAt VALUES ('T002', 2);
109 INSERT INTO ValidEntranceAt VALUES ('T003', 1);
110 INSERT INTO ValidEntranceAt VALUES ('T004', 3);
111 INSERT INTO ValidEntranceAt VALUES ('T005', 2);
112
113 -- 17. ValidExitAt
114 INSERT INTO ValidExitAt VALUES ('T001', 1);
115 INSERT INTO ValidExitAt VALUES ('T002', 2);
116 INSERT INTO ValidExitAt VALUES ('T003', 1);
117 INSERT INTO ValidExitAt VALUES ('T004', 3);
118 INSERT INTO ValidExitAt VALUES ('T005', 2);
119
120 -- 18. PassEnter
121 INSERT INTO PassEnter VALUES ('T001', 'G001', CURRENT_TIMESTAMP);
122 INSERT INTO PassEnter VALUES ('T002', 'G003', CURRENT_TIMESTAMP);
123 INSERT INTO PassEnter VALUES ('T003', 'G005', CURRENT_TIMESTAMP);
124 INSERT INTO PassEnter VALUES ('T004', 'G001', CURRENT_TIMESTAMP);
125 INSERT INTO PassEnter VALUES ('T005', 'G003', CURRENT_TIMESTAMP);
126
127 -- 19. PassExit
128 INSERT INTO PassExit VALUES ('T001', 'G002', CURRENT_TIMESTAMP);
129 INSERT INTO PassExit VALUES ('T002', 'G004', CURRENT_TIMESTAMP);
130 INSERT INTO PassExit VALUES ('T003', 'G002', CURRENT_TIMESTAMP);
131 INSERT INTO PassExit VALUES ('T004', 'G004', CURRENT_TIMESTAMP);
132 INSERT INTO PassExit VALUES ('T005', 'G002', CURRENT_TIMESTAMP);
```

AI Acknowledgment

Yes, Gemini was used as an AI tool during the preparation of this assignment. The full conversation with Gemini has been attached separately for reference and transparency.

Appendix A: Name Changes and Rationale

The following naming changes were made to improve consistency between the ER diagram, relational schema, and SQL DDL while following standard database naming conventions.

Original Name	Updated Name	Reason
LicensePlate	LicencePlate	Changed to match the spelling used consistently throughout the relational schema.
PassengerName	FirstName, LastName	Split into separate attributes to better represent passenger information and align with the ER diagram.
Fare	AmountPaid	More accurately describes the amount paid for a transit pass.
PurchaseDate	PurchaseTime	Changed to store the exact purchase timestamp instead of only the date.
ExpirationTime	StartTime, EndTime	Allows modelling passes with explicit validity periods.
ZoneNumber (ZonePass)	NumberOfValidZones	Clarifies that the attribute stores the number of valid zones rather than a specific zone.
GateType	EnterOrExit	Better reflects whether a gate is designated for entry or exit.

StopOrder	StopNumber	More descriptive and consistent with the route schema.
Status (Transport)	WheelchairAccessibility	Transport status was removed because operational status was outside the project scope; accessibility better reflects project requirements.
RouteDescription	RouteDescription / RouteDescription	Formatting difference only (line wrapping in the document); attribute name is unchanged.